

Binary Arithmetic Practice Sheet

Computer Architecture & Number Systems • Medium Level

25 Questions

1. When sign-extending an 8-bit 2's complement negative binary number 10110100 to 16 bits, what are the leading 8 bits?

- (A) 11111111 (B) 00000000
(C) 10110100 (D) 10000000

2. In an 8-bit 2's complement arithmetic system, an arithmetic overflow occurs when adding 01000000 (+64) and 01000001 (+65) because:

- (A) The sum exceeds +127 and flips the sign bit to 1 (B) There is a carry out of the MSB
(C) Both inputs are positive (D) The result becomes negative zero

3. What is the decimal value of the fractional binary number 0.1011_2 ?

- (A) 0.6875 (B) 0.7500
(C) 0.5625 (D) 0.6250

4. Subtract 110.1_2 from 1010.01_2 in binary arithmetic.

- (A) 0011.11 (B) 0100.01
(C) 0011.01 (D) 0100.11

5. What is the binary representation of decimal fraction 13.625?

- (A) 1101.101 (B) 1101.011
(C) 1100.101 (D) 1011.110

6. Perform an arithmetic right shift by 2 positions on the 8-bit 2's complement number 11110000 (-16).

- (A) 11111100 (-4) (B) 00111100 (+60)
(C) 11110000 (-16) (D) 11000000 (-64)

7. Overflow in signed 2's complement addition of two n -bit numbers A and B can be detected using carries C_{in} and C_{out} of the MSB as:

- (A) $V = C_{in} \oplus C_{out}$ (B) $V = C_{in} \cdot C_{out}$
(C) $V = C_{in} + C_{out}$ (D) $V = \overline{C_{in}} \oplus \overline{C_{out}}$

8. What is the product of two binary numbers 1.01_2 and 10.1_2 ?

- (A) 11.001 (B) 11.011
(C) 10.101 (D) 100.01

9. In an 8-bit system, what is the result of 01111111 (+127) + 00000001 (+1) in 2's complement representation?

- (A) 10000000 (-128, overflow) (B) 10000001 (-127, overflow)
(C) 01000000 (+64, valid) (D) 00000000 (0, no carry)

10. Which of the following ranges represents the numbers expressible in n -bit 1's complement form?

- (A) $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$ (B) -2^{n-1} to $+(2^{n-1} - 1)$
(C) -2^n to $+(2^n - 1)$ (D) $-(2^{n-1})$ to $+2^{n-1}$

11. Calculate 11011.011_2 divided by 100_2 .

- (A) 110.11011 (B) 11.011011
(C) 1101.1011 (D) 0.11011011

12. Convert 8-bit 2's complement number 10101100 to its decimal equivalent.

- (A) -84 (B) -172
(C) +172 (D) -83

13. Adding end-around carry is required in addition performed using:

- (A) 1's complement arithmetic (B) 2's complement arithmetic
(C) Sign-magnitude system (D) Unsigned binary addition

14. What happens when you perform XOR between a binary vector A and a vector of all 1s ($111\dots 1$)?

- (A) Yields 1's complement of A (B) Yields 2's complement of A
(C) Yields A unchanged (D) Yields all zeros

15. Compute $(-15) + (-20)$ using 8-bit 2's complement arithmetic. What is the resulting bit sequence?

- (A) 11011101 (B) 11011100
(C) 10100011 (D) 11100001

16. If a register stores 11010000 in sign-magnitude format (8-bit), what decimal number does it represent?

- (A) -80 (B) -48
(C) -208 (D) +208

17. Express decimal -0.625 as a normalized 8-bit binary fraction where MSB is the sign bit (1 for negative) followed by 7 fractional magnitude bits.

- (A) 11010000 (B) 10101000
(C) 01010000 (D) 11100000

18. Multiplying a 2's complement binary number by 4 is equivalent to:

- (A) Arithmetic left shift by 2 positions (B) Logical right shift by 2 positions
(C) Adding 0100 to the least significant bits (D) Inverting bit 2 and bit 3

19. Find the quotient and remainder when 11101_2 is divided by 101_2 in binary integer arithmetic.

(A) Quotient = 101, Remainder = 100

(B) Quotient = 110, Remainder = 001

(C) Quotient = 100, Remainder = 011

(D) Quotient = 101, Remainder = 010

20. In 8-bit 2's complement representation, what is the result of $(-128) + (+127)$?

(A) 11111111 (-1)

(B) 00000001 (+1)

(C) 10000000 (-128)

(D) 00000000 (0)

21. What is the representation of negative zero (-0) in 8-bit 1's complement format?

(A) 11111111

(B) 00000000

(C) 10000000

(D) 01111111

22. Compute $101.11_2 + 11.011_2$.

(A) 1001.001

(B) 1000.101

(C) 1010.010

(D) 1001.101

23. In binary restored division, if the remainder after subtraction is negative, the partial remainder is:

(A) Restored by adding the divisor back

(B) Shifted left without modification

(C) Converted to 2's complement

(D) Set to zero immediately

24. Subtracting a larger positive binary integer from a smaller positive binary integer yields a borrow out of MSB. In 2's complement arithmetic, this result is:

(A) Correct negative value in 2's complement

(B) Invalid due to unsigned underflow

(C) Equivalent to zero

(D) Positive number needing sign inversion

25. Calculate the bitwise operation $(10110100 \text{ AND } 11001100) \text{ OR } 00001111$.

(A) 10001111

(B) 11001111

(C) 10111111

(D) 10000100